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20.8.26

Setting up a local Area Network.

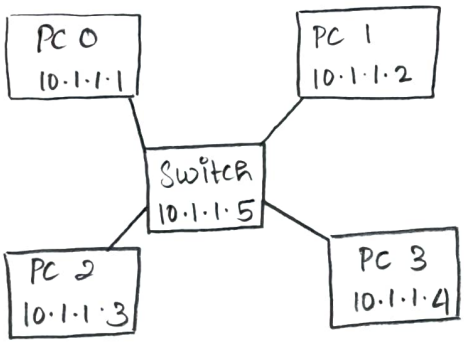
AIM:

To design, setup and configure a local Area Network (LAN) connecting four computers through a network switch, assign static IP addresses to all devices, verify connectivity using the ping command, and demonstrate resource / data storing across the network.

THEORY:

The local Area Network (LAN) is a network that connects computers and devices within a limited geographical area.

NETWORK DIAGRAM:



IP CONFIGURATION TABLE:

Device	IP Address	Subnet Mask	Connected to
PC 0	10.1.1.1	255.0.0.0	Switch port Fa0/1
PC 1	10.1.1.2	255.0.0.0	Switch port Fa0/2
PC 2	10.1.1.3	255.0.0.0	Switch port Fa0/3
PC 3	10.1.1.4	255.0.0.0	Switch port Fa0/4
Switch (management)	10.1.1.5	255.0.0.0	VLAN 1 Interface

Implementation in Cisco Packet Tracer:

Since physical hardware was limited, the LAN was also implemented and verified using Cisco packet Tracer simulation software, following the below steps.

Step 1: Place the Devices:

* Open cisco packet tracer
* From the device category panel, select end devices and drag 4 PCs onto the workspace.
* Select switches and drag 1 switch (eg: 2960 series) onto the workspace.

Step 2: Cable the Devices

* Click connections → select copper straight through cable.
* Click each PC, then click the switch and select an available fast Ethernet port to complete each connection.
* Repeat for all four PCs (Ports Fa0/1 to Fa0/4) Green link lights confirm a successful physical connection.

Step 3: Assign IP addresses to PCs

* Click PC → Desktop tab → IP configuration
* Enter the static IP Address and Subnet as per the IP table (section 5) for each PC

Step 4: Configure the switch management IP (optional)

* Click the switch → CLI tab
* Enter the following commands to assign a management.

IP:

```
enable
Configure terminal
interface vlan 1
ip address 10.1.1.5 255.0.0.0
no shutdown
exit.
```

Step 5: Verify connectivity using Ping

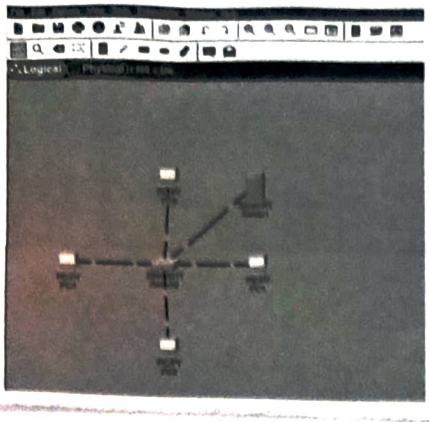
* Click a PC → Desktop → command Prompt
* Use the ping command to test connectivity to the other PCs Eg: ping 10.1.1.2
* Successful "Reply from..." messages confirm connectivity.

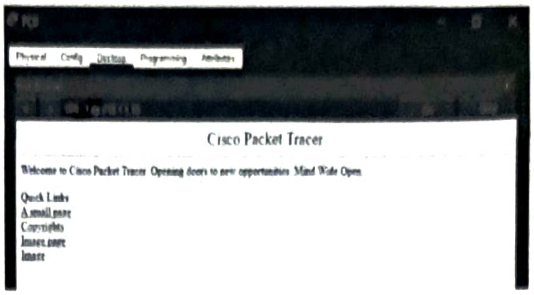
Step 6: Demonstrate Data Sharing (server simulation)

* Add a server device (End devices → Server PT) and connect it to the switch.
* Use the ping command to test connectivity to the ~~other~~ PCs switch.
* Assign it an IP address (Eg: 10.1.1.10, subnet mask 255.0.0.0) via its desktop → IP configuration tasks.
* Open the server's services (config) tab and enable HTTP service.
* From another PC, open the web browser (Desktop tab) and enter the server's IP address to load the page confirming successful data transfer across the LAN.

Step 7: Observe Packet flow in simulation mode.

* Send a ping or web request between two devices and use the play/capture controls to step through the packet as it travels hop by hop through the switch observing MAC addresses based forwarding.





RESULT:

Thus the setting up of a Local Area Network was observed.